```
import tensorflow as tf
from tensorflow.keras.datasets import imdb
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense
from tensorflow.keras.preprocessing.sequence import pad_sequences
import matplotlib.pyplot as plt
```

```
vocab_size = 10000
(X_train, y_train), (X_test, y_test) = imdb.load_data(num_words=vocab_size)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz
17464789/17464789 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
max_length = 200
X_train = pad_sequences(X_train, maxlen=max_length, padding='post')
X_test = pad_sequences(X_test, maxlen=max_length, padding='post')
```

```
model = Sequential()
model.add(Embedding(input_dim=vocab_size, output_dim=32, input_length=max_length))
model.add(LSTM(units=32))
model.add(Dense(units=1, activation='sigmoid'))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:100: UserWarning: Argum
  warnings.warn(
```

```
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

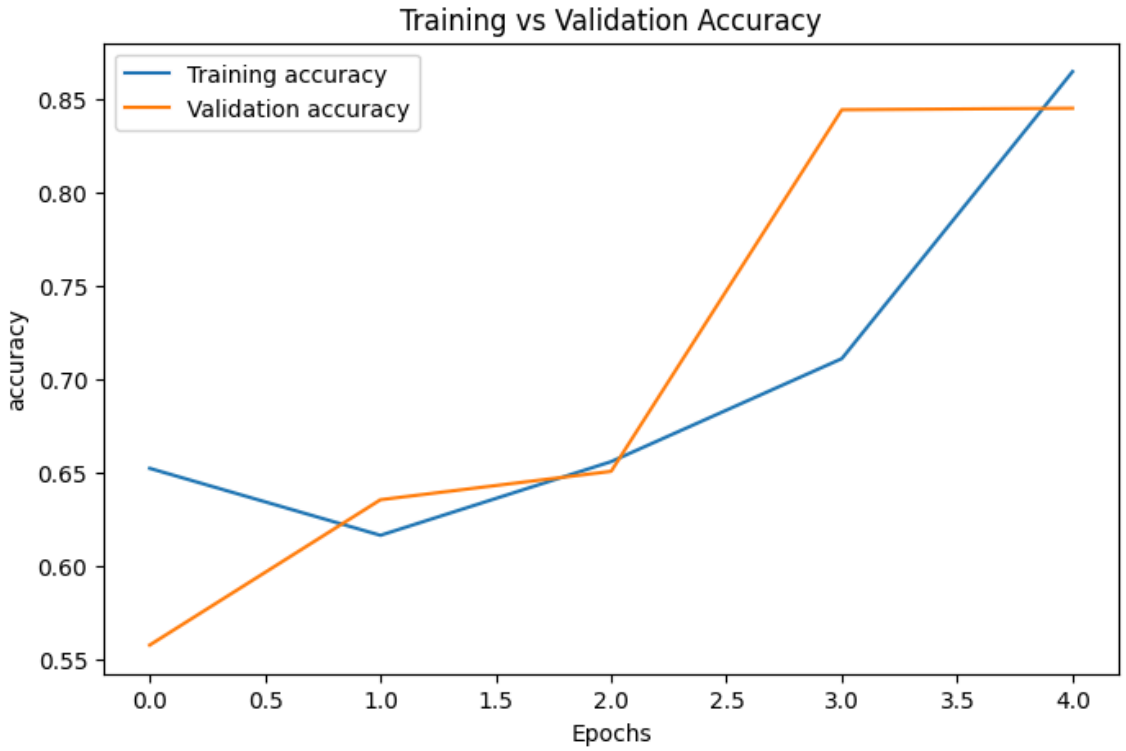
```
history = model.fit(X_train, y_train, epochs=5, batch_size=64, validation_split=0.2)
```

```
Epoch 1/5
313/313 ━━━━━━━━━━━━━━━━━━━━ 18s 58ms/step - accuracy: 0.6524 - loss: 0.6209 - val_accuracy: 0.557
Epoch 2/5
313/313 ━━━━━━━━━━━━━━━━━━━━ 18s 56ms/step - accuracy: 0.6166 - loss: 0.6383 - val_accuracy: 0.635
Epoch 3/5
313/313 ━━━━━━━━━━━━━━━━━━━━ 18s 58ms/step - accuracy: 0.6560 - loss: 0.5537 - val_accuracy: 0.650
Epoch 4/5
313/313 ━━━━━━━━━━━━━━━━━━━━ 20s 57ms/step - accuracy: 0.7111 - loss: 0.4954 - val_accuracy: 0.844
Epoch 5/5
313/313 ━━━━━━━━━━━━━━━━━━━━ 19s 52ms/step - accuracy: 0.8648 - loss: 0.3614 - val_accuracy: 0.845
```

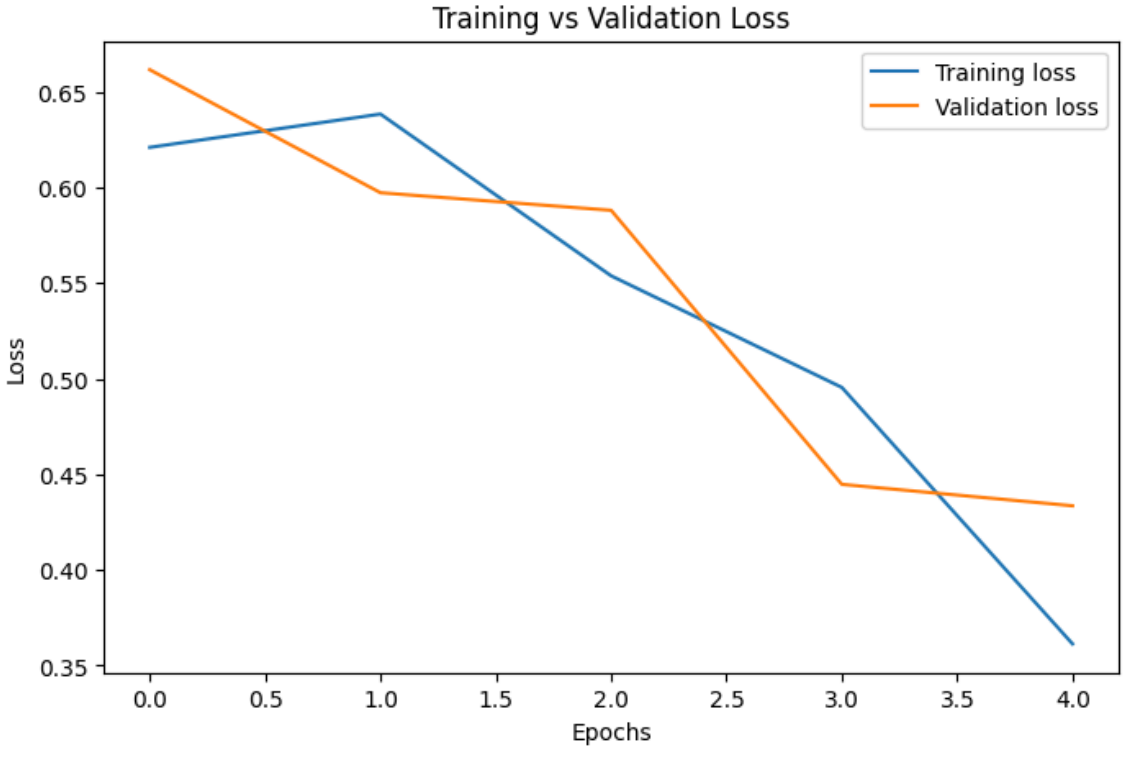
```
loss, accuracy = model.evaluate(X_test, y_test)
print("Test Loss:", loss)
print("Test Accuracy:", accuracy)
```

```
782/782 ━━━━━━━━━━━━━━━━━━━━ 9s 11ms/step - accuracy: 0.8390 - loss: 0.4433
Test Loss: 0.4433176517486572
Test Accuracy: 0.8389599919319153
```

```
plt.figure(figsize=(8, 5))
plt.plot(history.history['accuracy'], label='Training accuracy')
plt.plot(history.history['val_accuracy'], label='Validation accuracy')
plt.xlabel('Epochs')
plt.ylabel('accuracy')
plt.title("Training vs Validation Accuracy")
plt.legend()
plt.show()
```




```
plt.figure(figsize=(8, 5))
plt.plot(history.history['loss'], label='Training loss')
plt.plot(history.history['val_loss'], label='Validation loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.title("Training vs Validation Loss")
plt.legend()
plt.show()
```




```
predictions = model.predict(X_test[:10])
for i, prediction in enumerate(predictions):
    print(f"Review {i + 1}")
    print("Predicted Probability:", prediction[0])
    if prediction >= 0.5:
```

```
        print("Predicted Sentiment: Positive")
    else:
        print("Predicted Sentiment: Negative")
    print("Actual Label:", y_test[i])
    print("-"*40)
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 130ms/step
Review 1
Predicted Probability: 0.23114024
Predicted Sentiment: Negative
Actual Label: 0
----------------------------------------
Review 2
Predicted Probability: 0.9863108
Predicted Sentiment: Positive
Actual Label: 1
----------------------------------------
Review 3
Predicted Probability: 0.93211883
Predicted Sentiment: Positive
Actual Label: 1
----------------------------------------
Review 4
Predicted Probability: 0.23122111
Predicted Sentiment: Negative
Actual Label: 0
----------------------------------------
Review 5
Predicted Probability: 0.82375544
Predicted Sentiment: Positive
Actual Label: 1
----------------------------------------
Review 6
Predicted Probability: 0.23114036
Predicted Sentiment: Negative
Actual Label: 1
----------------------------------------
Review 7
Predicted Probability: 0.92770624
Predicted Sentiment: Positive
Actual Label: 1
----------------------------------------
Review 8
Predicted Probability: 0.22896773
Predicted Sentiment: Negative
Actual Label: 0
----------------------------------------
Review 9
Predicted Probability: 0.82350147
Predicted Sentiment: Positive
Actual Label: 0
----------------------------------------
Review 10
Predicted Probability: 0.9745127
Predicted Sentiment: Positive
Actual Label: 1
----------------------------------------
```